

AIAB Spring 2026 Coursework Assignment

Navigating the Reality Gap: Evolutionary Adaptations to
Noisy Communication in Bio-Inspired Swarms

Candidate number: 293275

10th May 2026

1 Introduction

1.1 Project Origins and Objectives

The study of swarm intelligence in biological systems provides a robust blueprint for the development of decentralized robotics (Bonabeau, Dorigo and Theraulaz, 1999). First formalized by Beni and Wang (1993), the field of swarm robotics is characterized by emergent intelligence produced through local interactions between computationally simple agents. This paradigm stands in direct contrast to traditional ‘monolithic’ robotics or GOFAI, which relies on a singular, highly intelligent agent or a centralized ‘brain’ to coordinate tasks. By distributing control, swarm systems achieve a level of dynamicity and fault tolerance essential for operating in volatile, real-world environments. Following these principles, this research investigates the ‘Waggle Dance’ of the honeybee (*Apis mellifera*) as a primary model for inter-agent communication within a simulated environment. Honeybee colonies were a clear choice for this investigation as they effectively integrate thousands of individual foragers into a coherent, higher-order entity through decentralized feedback loops (Seeley, 1995). While social recruitment in honeybees is optimized for natural environments, the transition of these biological rules into robotic systems often encounters the ‘Reality Gap’ - the discrepancy between abstract mathematical models and physical implementation (Jakobi, Husbands and Harvey, 1995). In this project, the honeybee foraging system is used as a lens to evaluate the trade-off between social communication and foraging efficiency in a simulated swarm. Specifically, the experiment tests whether inter-agent communication remains an asset or becomes a hindrance to collective performance when subjected to environmental noise.

1.2 Research Questions and Hypotheses

- RQ1: How does Gaussian noise in spatial communication affect collective foraging efficiency?
- RQ2: At what threshold of task precision does social recruitment transition to a metabolic liability?
- H1: In high-tolerance environments (0.8m patches), the GA will favor high Dance Probability and Recruitment Radii, as the Reality Gap is mitigated by target size.
- H2: In high-precision environments (0.2m patches), the swarm will prioritize independent scouting over social recruitment as the Signal-to-Noise ratio decreases.

1.3 Theoretical Framework

Swarm intelligence operates on the exploration-exploitation trade-off. While social recruitment accelerates resource acquisition (Seeley, 1995), it relies on high-fidelity information transfer. This is the framework under which this research operates in order to analyse at what point social recruitment in swarm algorithms is rendered a liability. Pybullet provides a Newtonian simulation environment by employing the Articulated Body Algorithm for linear-time forward dynamics and a unified LCP constraint solver (Coumans & Bai, 2016). This provides a grounded framework for this study to model the Reality Gap as stochastic interference, rather than abstract data corruption.

2 Methodology

2.1 Physics and Initial Conditions

The simulation was developed using the PyBullet physics engine, chosen for its ability to simulate rigid-body dynamics and provide a physically grounded alternative to abstract grid-world simulations (Couzin et al., 2005). To maintain computational stability, the simulation time step was fixed at $dt = 0.01s$, ensuring continuous collision detection. Gravity (*Code Listing 1, 020*) was set to $g = 9.81m/s^2$ (Z-axis), requiring agents to apply consistent lift forces to remain airborne. To ensure statistical significance and mitigate the effects of stochastic noise, each experimental scenario was conducted over 10 independent trials. Resources were structured as 6 discrete patches placed between 15–25m from the central hive. Implemented in *Code Listing 1, 062*, this distance was chosen empirically for its ability to eliminate sheer luck whilst maintaining possibility for discovery by the swarm. Each trial ran for a fixed duration of 120s (*Code Listing 1, 070*), a value determined to provide sufficient time for the swarm to complete multiple foraging, recruitment, and return cycles. Environmental consistency was maintained using a fixed seed (42) for debugging (*Code Listing 1, 061*), while trial variance was handled through the multi-run aggregation.

2.2 Agent Architecture: FSM

Each of the 100 agents operates via an asynchronous Finite State Machine (FSM) (*Code Listing 1, 098-187*). This architecture was selected because FSMs provide a lightweight, deterministic framework for handling reactive behaviors, which is a standard approach in swarm robotics to demonstrate emergent phenomena from simple, decentralized transitions (Bonabeau, Dorigo and Theraulaz, 1999). Movement control is governed by a proportional control law:

$$F_{steer} = k(pos_{target} - pos_{agent})$$

The proportional gain, $k = 0.6$, was selected via iterative tuning. Values exceeding 0.7 triggered aggressive overshoot and oscillations, while values below 0.5 resulted in sluggish tracking that failed to account for environmental friction. This balance is critical, as it ensures agents are responsive enough to follow the recruitment signal but stable enough to maintain a predictable flight path.

Algorithm 1 FSM State Transition Logic

```

1: State 0 (Searching):
2: if Memory is empty then
3:   Perform random walk (apply force  $F$ )
4: else
5:   Steer towards memory position
6: end if
7: if Collision with Pollen then
8:   Pollen counter ++, Save location  $\rightarrow$  State 1 (Returning)
9: end if
10: State 1 (Returning & Dancing):
11: if Agent is at Hive AND random value  $< P_{dance}$  then
12:   Broadcast memory location with noise  $\mathcal{N}(0, \sigma^2)$ 
13: end if
14: if Dance duration ends then  $\rightarrow$  State 0 (Searching)
15: end if
16: State 2 (Failure Recovery):
17: if Reached memory location AND no pollen found then
18:   MisguidedTrip counter ++, Clear mem  $\rightarrow$  State 0 (Searching)
19: end if

```

2.3 Evolutionary Strategy

The optimisation of swarm behaviour was conducted using a Genetic Algorithm (GA). This approach is justified by the ‘black-box’ nature of the search space, where an optimal relationship between noise and communication is unknown a priori (Mitchell, 1996). The GA optimizes three genes: $P_{dance} \in [0, 1]$, $T_{dance} \in [10, 500]$, $R_{recruit} \in [0.5, 10]$. The GA’s fitness is defined by $F_i = 10 \cdot P_{collected}$, and mutation by $Child = Child + \mathcal{N}(0, scale)$, ensuring genomic drift. Elite selection—retaining the top 20% of the population—was utilized to ensure that the most successful foraging strategies were preserved across generations, accelerating convergence (*Code Listing 1, 204-216*).

2.4 The Reality Gap

The Reality Gap, the phenomenon where controllers evolved in simulation fail to perform when transferred to physical hardware, remains a fundamental obstacle in evolutionary robotics (Jakobi, Husbands and Harvey, 1995). To model this, Gaussian noise was introduced into the agent’s communication channel, denoted by `reality_noise` in Code Listing 1, 221. The noisy coordinate (x_{noisy}) was derived from the true coordinate (x_{true}) as: $x_{noisy} = x_{true} + \mathcal{N}(0, \sigma^2)$. Here, $\mathcal{N}$ represents the normal distribution, $\mu = 0$ is the mean, and σ^2 is the variance (set to 0.8^2). This models non-systematic odometry error (Borenstein and Feng, 1996), creating a ”probability cloud” that recruits must navigate.

3 Results and Analysis

3.1 Analysis

Baseline Performance (Fig. 1(a)): The zero-noise control validated the Genetic Algorithm’s (GA) ability to evolve social recruitment in an idealized environment. By removing the ”Reality Gap,” the GA favored maximum dance probability ($P_{dance} \approx 0.6$), confirming H1. Peak efficiency reached 30 units, establishing an upper performance bound. Notably, the Recruitment Radius ($R_{recruit}$) decreased during this phase, suggesting that under perfect information fidelity, the swarm prioritizes precise communication over broad spatial reach.

High-Tolerance Environment (Fig. 1(b)): When noise ($\sigma = 0.8$) was introduced alongside a maintained $0.8m$ patch radius, P_{dance} remained volatile but sustained, indicating that the Signal-to-Noise Ratio (SNR) was sufficient to maintain social recruitment. To compensate for spatial uncertainty, the swarm evolved a higher $R_{recruit}$ compared to the baseline, effectively adopting a ”brute-force” strategy to increase the probability of recruits locating the patch despite coordinate drift.

Low-Tolerance Threshold (Fig. 1(c)): The transition to a $0.2m$ patch radius under constant $\sigma = 0.8$ noise resulted in a catastrophic collapse in foraging efficiency (≈ 9 units). The Reality Gap rendered the recruitment mechanism ineffective under these high-precision constraints. The high frequency of misguided trips (peaking at ≈ 900 in Gen 3) indicates that noisy coordinates consistently directed recruits to empty space. Although P_{dance} spiked mid-run, efficiency failed to recover, confirming H2. The ”metabolic cost” of following erroneous signals outweighed the benefit of social recruitment, effectively rendering the waggle dance a maladaptive strategy in low-tolerance environments.

Figure 1

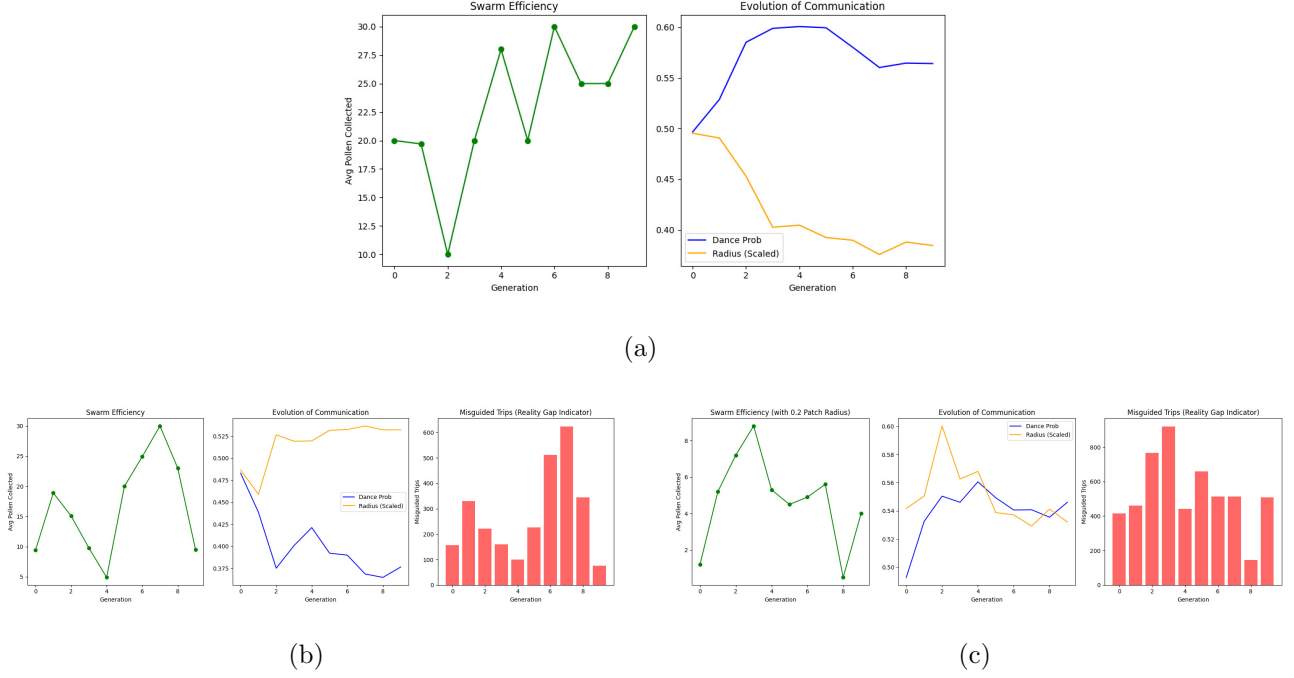


Fig 1: Figure 1(a) shows the 'baseline' experiment where communication evolves over 10 generations with no noise. Efficiency (pollen units collected) is depicted by the green line on the left whilst P_{dance} (blue) and R_{radius} (orange) are shown on the right. These graphs are used to analyse the relationship between communication and swarm efficiency. Figure 1(b) and 1(c) are structured in the same way with the addition of 'Misguided Trips' (red), depicting the effect of corrupted coordinates ($\sigma = 0.8$). All figures were generated from the results in Code Listings 2-4.

3.2 Quantitative Comparison Summary

Table 1: Quantitative comparison of swarm performance across experimental scenarios.

Scenario	Max Efficiency	Avg. Misguided Trips	Primary Evolved Strategy
No Noise	30.0	≈ 0	High-frequency social sharing
0.8m Patch (High Tol)	30.0	≈ 350	Robust communication; high radius
0.2m Patch (Low Tol)	9.0	≈ 550	Evolutionary instability; scouting favoured

4 Evaluation

4.1 Success in Addressing Objectives and Hypotheses

The primary objective of this research was to identify the threshold at which social communication becomes a “metabolic liability” due to the Reality Gap (SNR). The experimental results conclusively

answer RQ1 and RQ2, demonstrating that foraging efficiency is not just a product of communication frequency, but of the relationship between signal precision and target size. The results in Figure 1(b) support H1; in high-tolerance environments, the GA maintained high communication levels. Figure 1(c) provided strong evidence for H2 - as the target shrank to 0.2m, the mismatch between noise and task precision led to a collapse in efficiency. The high frequency of “misguided trips” proves that the swarm was trapped in a cycle of following incorrect information, verifying that communication can indeed hinder a swarm when noise is not accounted for (Jakobi, Husbands and Harvey, 1995).

4.2 Significance and Theoretical Implications

This study highlights a critical nuance in bio-robotics: biological rules, such as the honeybee waggle dance, are evolved for specific environmental constraints. As Seeley (1995) notes, honeybee colonies are optimised for natural resource distributions. The results suggest that when translating these biological rules in decentralised robotics, the fidelity of the channel is as important as the logic of the agent. The transition from social recruitment to independent scouting observed in the Low Tolerance trial suggests that a truly robust swarm must be capable of dynamic strategy - abandoning social cues when an environment becomes too volatile.

4.3 Limitations and Methodological Constraints

While the project was successful in addressing the Reality Gap, two primary limitations affect the clarity of the results:

1. Population Bottleneck: With a population of 100 over only 10 generations, the GA occasionally converged on local optima (seen in the sudden efficiency drops in Figure 1(a) and 1(c)). A more robust study would require 50-100 generations to see if the swarm eventually “evolves away” the dance probability entirely in high-noise scenarios.
2. Static Noise Profile: The Gaussian Noise was held at a constant $\sigma = 0.8$. Implementing a dynamic noise profile - where noise increases with distance from the hive - would have provided a more realistic model of the “Odometry Error” found in physical robots.

4.4 Future Works and Revised Objectives

If this research were to continue, the objectives should be revised to investigate Signal Filtering. Instead of simply reducing communication frequency, the agents could evolve a “skepticism” threshold:

- Revised Hypothesis: A swarm that utilises “Bayesian Filtering” to cross-reference social signals

with individual memory would outperform a purely “trusting” swarm in low-tolerance environments.

- Implementation: Future iterations would replace the FSM with a simple Neural Network (NN) to allow for more complex decision-making regarding when to ignore a dance (Mitchell, 1998).

Bibliography

Beni, G. and Wang, J. (1993). Swarm intelligence in cellular robotic systems. [online] Available at: <https://api.semanticscholar.org/CorpusID:59721358>.

Bonabeau, E., Dorigo, M. and Theraulaz, G. (1999). Swarm intelligence: from natural to artificial systems. [online] New York; Oxford: Oxford University Press. Available at: https://www.researchgate.net/profile/Guy-Theraulaz/publication/243779308_From_Natural_to_Artificial_Swarm_Intelligence/links/5f60f4cca6fdcc1164156587/From-Natural-to-Artificial-Swarm-Intelligence.pdf.

Borenstein, J. and Feng, L. (1996). Measurement and correction of systematic odometry errors in mobile robots. IEEE Transactions on Robotics and Automation, [online] 12, pp.869–880. doi:<https://doi.org/10.1109/70.544770>.

Coumans, E. and Bai, Y., (2016). Pybullet, a python module for physics simulation for games, robotics and machine learning [online] Available at: <https://docs.google.com/document/d/10sXEhzFRSnnvFc13XxNGhnD4N2SedqwdAvK3dsihxVUA/edit?tab=t.0>

Couzin, Iain D, Krause, J., Franks, N.R. and Levin, S.A. (2005). Effective leadership and decision-making in animal groups on the move. Nature (London), [online] 433, pp.513–516. Available at: https://sussex.primo.exlibrisgroup.com/discovery/fulldisplay?docid=cdi_pascalfrancis_primary_16469220&context=PC&vid=44SUS_INST:44SUS_VU1&lang=en&search_scope=MyInst_and_CI_no_BLDS&adaptor=Primo%20Central&tab=MyInst_and_CI_no_BLDS&query=any,contains,Couzin%202005&offset=0.

Jakobi, N., Husbands, P. and Harvey, I. (1995). ‘Noise and the reality gap: The use of simulation in evolutionary robotics’,. In: Lecture Notes in Artificial Intelligence. [online] pp.704–720. Available at: https://www.researchgate.net/publication/221530885_Noise_and_the_Reality_Gap_The_Use_of_Simulation_in_Evolutionary_Robotics.

Mitchell, M. (1996). An introduction to genetic algorithms. [online] Cambridge, Mass.: MIT Press. Available at: <https://doi.org/10.7551/mitpress/3927.001.0001?locatt=mode:legacy>.

Seeley, T.D. (1995). The wisdom of the hive : the social physiology of honey bee colonies. [online] Harvard University Press. Available at: <https://sussex.idm.oclc.org/login?url=https://ebookcentral.proquest.com/lib/suss/detail.action?docID=3300714>.